Fredrick Ondara

Electrical & Electronic Engineering Student | Telecom, Embedded Systems & Web Development

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PROFESSIONAL SUMMARY

Electrical and Electronic Engineering student seeking an internship, attachment or graduate engineering role in telecommunications. Certified in **Fiber Optics Technology**, with hands-on experience in embedded systems, IoT, automation and control simulation. Founder of Globyx, an online virtual number and proxy platform. Builds and ships real products, from robotics control systems to responsive web applications. A fast self-learner who works across hardware, software and cloud, and uses modern AI tools to speed up design, debugging and documentation.

EXPERIENCE

Founder

Globyx globyxlimited2026.vercel.app

- Founded and run an online platform offering virtual phone numbers in 60+ countries and residential, mobile, ISP and datacenter proxies.
- Platform gives customers a single wallet and one dashboard to manage their numbers and proxies.

Research Assistant

University Lab & Independent Projects

- Contributed to robotics, IoT and automation research.
- Built and tested control systems integrating MATLAB with CoppeliaSim simulations.
- Debugged hardware and firmware across Arduino and serial bus servo platforms.

Robotics Team Leader

University Automation Club

- Lead a student robotics team through planning, building and testing of automation projects.

FEATURED PROJECTS

MATLAB-CoppeliaSim Fusion
Developed fuzzy logic control for Pioneer P3DX robots using MATLAB integration and Lua scripting; built a real-time robotics control system.

Hiwonder Robotics Series (xArm1S & LX-225 Servos)
Created programmable robotic arm experiments using Arduino and Python.

Terminal-Bench 3 Project
Worked on a Terminal-Bench 3 project, a benchmark that tests AI agents on hard, realistic command-line tasks, developed with Claude Fable 5.1 as an AI coding agent.

Globyx globyxlimited2026.vercel.app
Personal business project: virtual phone numbers in 60+ countries and multi-type proxies, managed from one wallet and dashboard.

Epiculean Journeys epicurean-journeys.vercel.app
Built a responsive travel and cuisine website with modern UI/UX, deployed on Vercel.

Tales Untold tales-untold.vercel.app
Built a responsive storytelling website with modern UI/UX, deployed on Vercel.

EDUCATION

BSc Electrical & Electronic Engineering
Technical University of Mombasa, Kenya
Expected graduation: 2028

High School Certificate
Distinction in Mathematics, Physics and Computer Studies.

CERTIFICATIONS

- **Fiber Optics Technology** (Certified): optical communication systems and installation
- AWS Cloud: ongoing study of cloud computing
- Web Development: self-taught and project-based

TELECOM & NETWORKS

- Optical fiber systems
- IoT connectivity and cloud fundamentals

EMBEDDED & HARDWARE

- Arduino, ESP32
- FPGA programming
- Circuit and PCB design
- Sensor integration
- Serial bus servo debugging

SIMULATION & CONTROL

- MATLAB
- CoppeliaSim / V-REP
- Fuzzy logic control
- AutoCAD

PROGRAMMING & WEB

- C, C++, JavaScript/TypeScript, Python, Lua, HTML/CSS
- React, Tailwind CSS, Bootstrap, Framer Motion
- Git/GitHub, Figma, Arduino IDE

AI-ASSISTED ENGINEERING

- GPT, Gemini, Claude (Opus, Fable), Google Antigravity, Jules
- Multi-agent workflows for coding, research and documentation

AFFILIATIONS

- Active Member, IEEE Student Branch

SOFT SKILLS

- Communication and Team Collaboration
- Problem Solving and Time Management
- Self-Learning

LANGUAGES

- English (Fluent)
- Swahili (Fluent)

REFERENCES

- Dr Shitsukane Siliali: 0724 963 044
- Dr Charles Majani: 0725 670 585
- Certificates available on request